

# Leveraging Large Language Models for Contextual Prioritization of Contaminants of Emerging Concern in Chemical Mixtures

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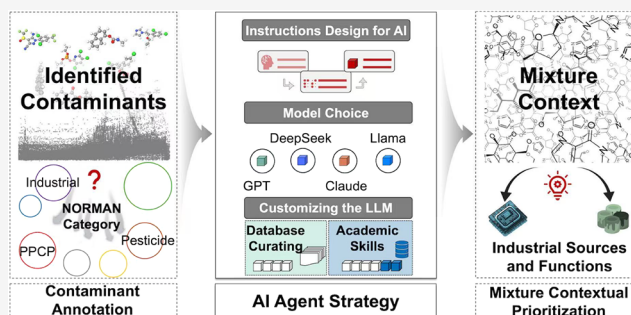
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**ABSTRACT:** Effective management of chemical mixtures presents a continuing challenge due to the growing diversity and inadequate characterization of contaminants of emerging concern (CECs). While recent advances in nontarget analysis enable the generation of extensive chemical inventories, key bottlenecks have shifted to postidentification interpretation within heterogeneous data. Here, we present an agent-based workflow that integrates large language models (LLMs) with functional categories, potential sources, and toxicology information to support risk prioritization. The practical technical components and evaluation benchmarks for LLMs were established, showing that optimized prompts and the best-performing model (GPT-4-Turbo) among the seven candidates enhanced user alignment with context perfectly. Integrating real-world data through retrieval-augmented generation enabled us to retrieve 100% truthful content, and further fine-tuning nearly doubled response consistency, substantially reducing hallucination. The workflow was validated using two mixture scenarios to assess the applicability across matrices and chemical contexts. The agent enabled complete functional and source annotation of chemicals by querying the NORMAN Network and achieved ~85% accuracy for substances absent from existing databases by emulating NORMAN-aligned logic. This capability allowed mixture-level interpretation of chemical inventory, revealing dominant categories and industrial sources, such as lubricants in shale gas flowback produced water and semiconductor-related industrial intermediates, which contributed to elevated risks in the studied scenarios.

**KEYWORDS:** AI agent, environmental risk assessment, large language model, artificial intelligence, chemical mixtures



## INTRODUCTION

Chemical mixtures are composed of diverse components, many of which remain unidentified, presenting decided challenges to environmental risk assessments.<sup>1</sup> To determine key risk drivers in chemical mixtures and their potential sources, advanced approaches are needed to broaden the scope of identifiable compounds while efficiently prioritizing high-risk substances.<sup>2–4</sup> Despite recent progress in detecting contaminants of emerging concern (CEC) in the environment through updated analytical paradigms and sophisticated instrumentation (e.g., high-resolution mass spectrometry, HRMS), the subsequent translation of this rapidly expanding chemical space into actionable, management-relevant information has not progressed at a comparable pace.<sup>5,6</sup> While such an improvement has boosted the ability to uncover nontarget CECs, many of these compounds may not merit prioritization in environmental studies. Therefore, to prevent inefficient allocation of resources in future management efforts, it is essential to advance postidentification measures that extend beyond detection alone. Such efforts support screening- and prioritization-oriented decision-making through the contextual interpretation of chemical categories, potential sources, exposure patterns, and toxicity profiles.<sup>7</sup> Strengthening this postidenti-

fication decision-support layer is critical to ensuring that monitoring, risk assessment, and regulatory actions remain focused on manageable subsets of pollutants most likely to contribute to mixture risks, rather than on the expanding chemical inventory itself.

Risk prioritization and source apportionment require integration and contextual interpretation of heterogeneous information related to CECs, rather than reliance on any single benchmark scheme. Functional categorization frameworks, such as those developed by the NORMAN Network (NN, <https://www.norman-network.com/nds/>)<sup>8,9</sup> have been widely adopted to support regulatory-oriented pollutant grouping and to facilitate communication across risk assessment and management domains. Importantly, the NN is designed as a pragmatic, management-oriented classification system instead of a comprehensive chemical ontology or predictive model. Its

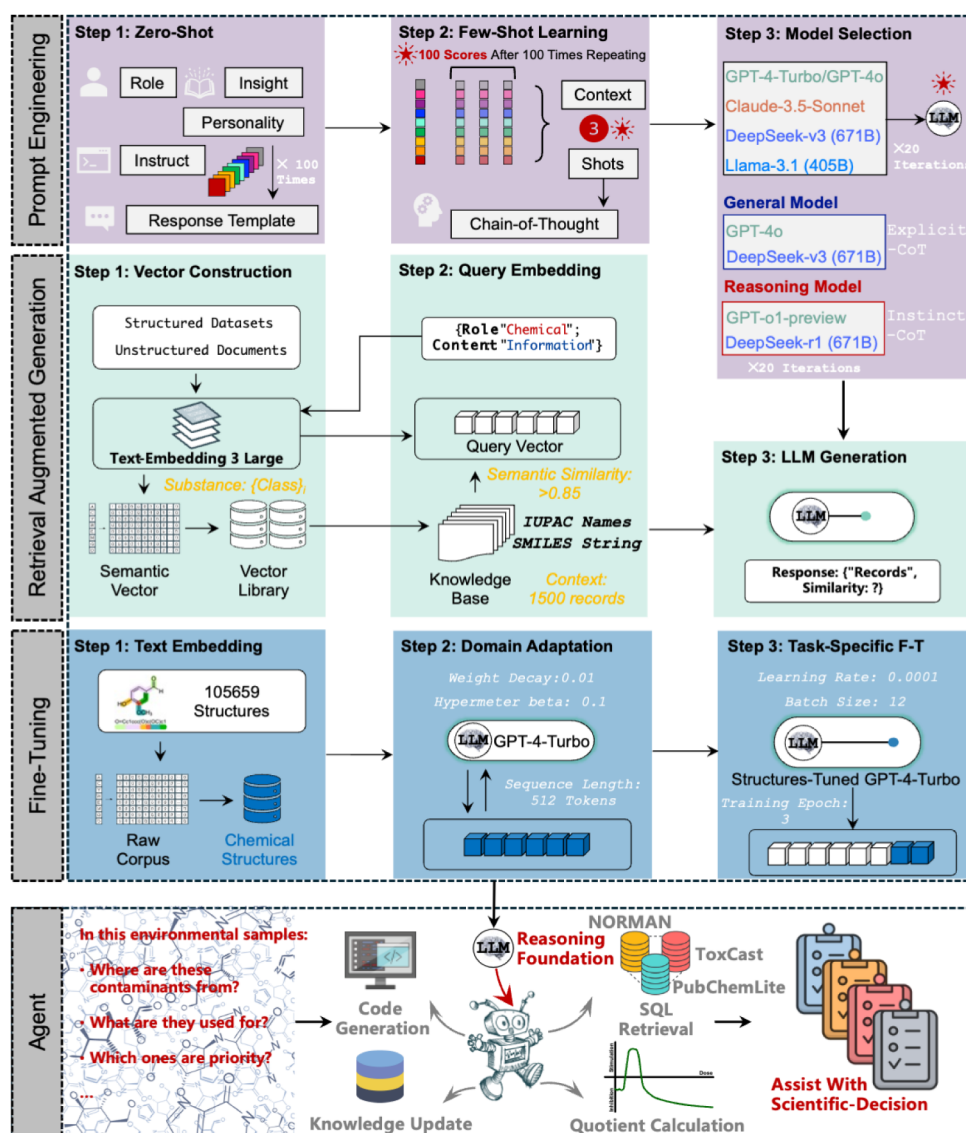
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**Figure 1.** Research framework for an AI agent-based large language models (LLMs) workflow to support prioritization-oriented scientific decision for contaminants of emerging concern (CECs) in chemical mixtures. LLM behavior is evaluated under progressively constrained settings: prompt engineering (in purple), retrieval-augmented generation (in green), and fine-tuning (in blue). In the agent module, LLMs are restricted to orchestration and interpretation roles by generating SQL queries to structured resources (e.g., NORMAN, PubChemLite, and ToxCast) and summarizing retrieved evidence into transparent, reviewable decision-support outputs, with optional user updates to the knowledge base.

key value lies in providing a structured entry point for subsequent prioritization. However, as the chemical space of identifiable CECs continues to expand, risk-informed prioritization increasingly depends on the systematic analysis of complementary evidence. In practice, this demands integrating information from multiple structured resources, including databases focused on chemical properties and identifiers (e.g., PubChem<sup>10</sup>), exposure and toxicology data (e.g., CompTox Chemicals Dashboard<sup>11</sup>), and bioactivity evidence (e.g., Comparative Toxicogenomics Database<sup>12</sup>). At present, such integration relies largely on manual database searches, which often yield fragmented and unprocessed outputs that are difficult to translate into coherent, decision-relevant insights. In contrast, artificial intelligence (AI)-assisted data integration offers a promising alternative by coordinating and contextualizing information retrieved from multiple databases. Previous studies have demonstrated the value of AI-driven approaches for extracting and organizing environmental knowledge, for

instance, using text mining to extract structural information from large document corpora,<sup>13</sup> constructing relational databases to bridge disparate data types,<sup>14</sup> and applying machine learning to uncover latent patterns or associations across datasets.<sup>15</sup> Nevertheless, many of these solutions remain difficult to operationalize in routine environmental practice, as high technical barriers, methodological fragmentation, and limited interpretability hinder their accessibility to non-specialist users.<sup>16</sup>

Recent advances in LLMs, including GPT and DeepSeek, have attracted global attention for their generative applications across specialized professional tasks and knowledge integrations.<sup>17–19</sup> In environmental science, LLMs have shown potential in automating the extraction of antioxidants, thereby facilitating the development of cost-efficient toxicological databases.<sup>20</sup> The GPT was also used for organizing and querying multilingual policy documents regarding antimicrobial resistance, illustrating its ability to help bridge knowledge

gaps in global environmental health governance.<sup>21</sup> These studies have shown the potential of LLMs to support environmental data integration. Unlike conventional machine learning models, LLMs are trained on vast, robust datasets containing billions of parameters, which confers an exceptional capacity for few-shot learning, the ability to discern complex task patterns under minimal supervision.<sup>22,23</sup> However, when LLMs are applied to specialized scientific domains without explicit constraints, their flexibility compromises reliability, leading to hallucination, manifested as misinterpretation of instructions, factual inaccuracies, or internally inconsistent outputs, which strongly limit their applicability.<sup>24,25</sup> In environmental science practice, decision-making often depends on traceable evidence and well-defined regulatory logic; thus, unconstrained generative reasoning is insufficient and can even be misleading. To address these challenges, a variety of strategies has been proposed,<sup>26</sup> including prompt engineering<sup>27</sup> with chain-of-thought (CoT) reasoning,<sup>28,29</sup> the use of reasoning-optimized architectures<sup>30</sup> and retrieval-augmented generation (RAG)<sup>31</sup> to dynamically query external knowledge bases,<sup>32</sup> and fine-tuning (F-T) with domain-specific data.<sup>33,34</sup> Although these efforts can reduce hallucination and improve task performance, their effectiveness depends critically on how LLMs are embedded within a broader workflow, which constrains their role and defines their interaction with structured data sources. In environmental applications, especially, treating LLMs as autonomous classifiers or predictors remains problematic. These considerations motivate a shift from viewing LLMs as independent analytical tools toward their integration as components within agent-based decision-support systems. In such systems, structured chemical, exposure, and toxicological databases serve as authoritative information sources, while LLMs function as coordination and interpretation layers.

Accordingly, the present study evaluated the use of LLMs as copilot components within an AI agent-based framework for environmental risk prioritization. Environmentally relevant categorization is used as a practical and quantifiable entry point to assess how an LLM, when provided with standard chemical identifiers (e.g., IUPAC names or SMILES), can support screening- and prioritization-oriented decision-making by coordinating and interpreting information retrieved from structured environmental and toxicological databases. We developed an agent-based workflow that integrates prompt design, RAG, domain-specific F-T, and the agent platform to constrain the LLM to a coordinating and interpreting role. Furthermore, we examined the performance of several state-of-the-art LLMs and their reasoning-oriented variants to evaluate the robustness and consistency under this risk management-oriented setting using real-world environmental samples.

## METHODS

As shown in Figure 1, we developed an agent-based, LLM-assisted environmental decision-support workflow. Environmentally relevant categorization was used as a constrained entry point for postidentification screening and prioritization. Model performance was evaluated using a dataset of 1104 CECs, comprising 623 compounds in shale gas flowback produced water (FPW)<sup>35</sup> and 481 in urban sediment samples from the Pearl River, China<sup>36</sup> identified by HRMS-based nontarget analysis (NTA). These CECs represented real-world chemical mixtures from distinct environmental matrices in a postidentification scenario. Unlike conventional machine learning, which trains and validates models from scratch, the present study focused on reasoning, task adaptation, and reasoning control using

pretrained LLMs within an agent-based framework, with performance progressively constrained through prompt engineering, RAG, and F-T. The workflow is detailed below. All codes and data used are available at <https://github.com/fabercheng/LLM>.

## Data Compilation

A comprehensive category knowledge base was compiled by aggregating 83,152 substances from NN (accessed on March 21, 2024; [Supplementary Sheet 01](#)), each documented with IUPAC names, SMILES structures, and categorical information. Prior to LLM development, a classification scheme was predefined following the NN taxonomy, with minor modifications. According to functional and source similarity, the original NN categories were merged into nine categories ([Figures 1 and S1](#)). This modification also involved removing several NN categories to mitigate potential classification ambiguities and overlaps. In addition, bioactivity data were acquired from InvitroDB v4.3 (accessed on October 28, 2025) and prepared as MySQL to support toxicological context retrieval within the agent framework. Exposure- and usage-related information was obtained from the PubChemLite for Exposomics (v20250802, accessed on October 28, 2025), which was downloaded and curated locally as a MySQL file to enable structured query and integration during agent-based reasoning.

To support LLM reasoning, we also compiled a small dataset containing 1104 NTA-identified compounds ([Supplementary Sheet 02](#)). A manual search indicated that 313 of the FPW-related compounds were listed in the NN database, providing established reference categorizations ([Supplementary Sheet 03](#)). These 313 substances were used as the reasoning set for prompt engineering and benchmarking throughout the entire workflow. Meanwhile, 413 compounds from both matrices, which were absent from NN, were assigned to the predictive set of optimized LLM strategy, and their categorizations were acquired through manual curation using online sources ([Supplementary Sheet 04](#)). For all substances, IUPAC names and SMILES structures were retrieved from PubChem and CCD, and MACCS molecular fingerprints were generated using RDKit based on canonical SMILES.

## Prompt Engineering

Four strategies were applied to optimize the LLM performance for categorization tasks using the reasoning set of 313 compounds ([Figure 1](#)). First, a series of zero-shot learning prompts were designed, incorporating multiple elements such as role definition, insight prompts to elicit domain-specific reasoning based on environmental knowledge, personality configuration to align the model's response style with professional expectations, emphasized instructions with symbolic cues (e.g., bolded keywords, exclamation marks, bracketed sections) to highlight task-critical elements, and a JSON response template. Second, a human feedback-based supervised process was implemented. Using this reasoning data, a Q&A format was established for each substance, where 100 questions were queried and 100 corresponding answers were generated. Classification results of each answer were collected and statistically analyzed. Third, few-shot learning prompts were developed to further boost the model performance. For each of the nine categories, three representative shots were included in the prompt. These shots were selected according to manual check results under human feedback, particularly the three highest-scoring answers, where the model correctly classified the substance 100 times out of 100. Fourth, a multilanguage prompt was engineered by integrating natural language descriptors (IUPAC names) and machine-readable chemical notations (SMILES and MACCS fingerprints). Prompt templates for all strategies are provided in the [Supporting Information](#).

## Model Selection

To identify the most effective LLMs for the contextual categorization task, we evaluated seven state-of-the-art models with distinct architectures and reasoning capabilities. Six general-purpose LLMs, including GPT-4-Turbo, GPT-4o, GPT-5, Claude-3.5-Sonnet, DeepSeek-v3 (671B), and LLaMA-3.1 (405B), were examined by using explicit CoT prompting, for which reasoning was guided by human



designed templates. In comparison, we also evaluated two reasoning-optimized LLMs, GPT-o1-preview and DeepSeek-r1 (671B), using instinctive CoT, which requires minimal instructions and allows the model to reason classification logic autonomously. Performance benchmarking between general and specialized architectures was also conducted: GPT-4o versus GPT-o1 and DeepSeek-v3 versus DeepSeek-r1. All LLMs were deployed with default settings for the context length, maximum response tokens, and decoding parameters. To ensure deterministic and reproducible outputs, the temperature was fixed at zero.

### Retrieval-Augmented Generation (RAG) and Fine-Tuning (F-T)

To improve classification performance, RAG and supervised F-T were sequentially applied to GPT-4-Turbo and GPT-5. The RAG incorporated domain-specific knowledge through embeddings of over 80,000 substances from the NN database, and IUPAC names and SMILES strings were used as chemical identifiers in the “Text-Embedding 3 Large” model to construct a semantic vector library. Query inputs were similarly embedded and matched using cosine similarity (threshold = 0.85, max retrieval = 1,500), enabling LLM access to domain-specific knowledge. When there was no match, predictions relied on pretrained structural understanding. The subsequent F-T used an embedded structured dictionary containing SMILES, IUPAC names, and categories to fine-tune model parameters (learning rate = 0.0001; batch size = 32; 3 epochs). Prompt templates defined role and category targets to improve classification consistency and reduce hallucination. The resulting “pollutant-tuned GPT-4-Turbo” showed good classification performance for CECs with and without NN classification labels. More details on vector library construction and F-T corpus preprocessing are presented in the [Supporting Information](#).

### Agent Platform

To enable end-to-end LLM reasoning execution under constrained and reproducible settings, we constructed an agent platform that integrates FastGPT, FDA, and Dify as complementary functional layers. FastGPT (<https://github.com/labring/FastGPT>) was utilized to manage the domain-specific knowledge bases and to deploy optimized LLM configurations, including the ingestion and updating of structured chemical, exposure, and toxicological resources used for retrieval and reasoning. FDA (<https://github.com/kangarooking/fastgpt-dify-adapter>) was used to establish and manage a set of standardized application programming interfaces (APIs) that support modular access to external databases and analytical services during agent execution. Dify (<https://github.com/langgenius/dify>) was used to assemble the agent workflow, defining the task sequence, conditional logic, and information flow among the retrieval, reasoning, and output modules. Specifically, it was used to generate structured SQL queries to interact with relational databases during agent execution. These platforms are accessible via [https://huggingface.co/Qianhui19/CECs\\_annotating\\_agent](https://huggingface.co/Qianhui19/CECs_annotating_agent).

### Reasoning Evaluation

A “Golden Rule” evaluation framework was applied to LLM reasoning, in which the reasoning set of 313 substances had classification labels accessed from the NN database. This setup enabled evaluation of LLM reasoning following few-shot learning optimization. Initial validation focused on prompt engineering optimization through repeated Q&A sessions for each substance, and classification results were compared against the NN database reference values. Model selection procedures used independent experimental replicates to account for inherent variability in LLM outputs. For RAG and F-T procedures, each substance was subjected to repeated Q&A sessions. Although different repetition schemes were implemented across evaluation steps, statistical analysis confirmed that these differences did not introduce significant variance in performance metrics ([Figure S2](#)). Model performance was assessed with three benchmarks: User alignment, Truthfulness, and Consistency. Three benchmarks were developed following OpenAI’s Instruct-GPT framework, which leverages reinforcement learning

from human feedback (RLHF).<sup>26</sup> Besides, these three benchmarks were consistent with core dimensions proposed in the TrustLLM evaluation framework.<sup>37</sup> Prompt design instructed the model to output one primary category (representing the most likely class within the study’s context) and up to two secondary categories (alternative plausible classifications). While Consistency was based on the correct option appearing randomly in the three categories, User alignment and Truthfulness were only referred to the top-ranked primary one.

User alignment examines whether the LLM selects a classification outside the nine predefined categories in the categorization task. For a given substance, assuming that the classification provided by the model is  $C_i$  and the nine predefined categories are  $\{C_1, C_2, \dots, C_9\}$ , UA is defined following [eq 1](#).

$$\text{User alignment} = \frac{\sum_{i=1}^N \delta(C_i \in \{C_1, C_2, \dots, C_9\})^{(j)}}{N}, j = 1, 2, \dots, M \quad (1)$$

where  $\delta$  is an indicator function, which equals 1 if  $C_i$  is one of the nine categories and 0 otherwise;  $N$  is the number of responses for each substance; and  $M$  is the total number of substances in the reasoning set.

Truthfulness examines whether the model output of classification results is consistent with the NN classification labels. For each substance, the true classification from NN is  $T_i$ , and the model-predicted classification is  $P_i$ . Truthfulness is defined following [eq 2](#).

$$\text{Truthfulness} = \frac{\sum_{i=1}^N \delta(T_i = P_i)^{(j)}}{N}, j = 1, 2, \dots, M \quad (2)$$

where  $\delta$  is an indicator function that equals 1 if the model prediction  $T_i$  matches the true classification  $P_i$  and 0 otherwise;  $N$  is the number of responses for each substance; and  $M$  is the total number of substances in the reasoning set.

Consistency evaluates model stability and reproducibility by analyzing response patterns across multiple trials for each substance and was calculated using [eq 3](#).

$$\text{Consistency} = \frac{\sum_{i=1}^N \delta(P_i = P^*)^{(j)}}{N}, j = 1, 2, \dots, M \quad (3)$$

where  $\delta$  is an indicator function that equals 1 if the model prediction  $P_i$  matches the most frequent answer across multiple responses  $P^*$  and 0 otherwise;  $N$  is the number of responses for each substance; and  $M$  is the total number of substances in the reasoning set.

To facilitate model ranking, we calculated an integrative Score as the weighted arithmetic mean of the three benchmarks mentioned above ([eq 4](#)).

$$\text{Score} = (\omega_1 \cdot \text{User alignment} + \omega_2 \cdot \text{Truthfulness} + \omega_3 \cdot \text{Consistency}) \times 100 \quad (4)$$

where weights  $\omega_1$ ,  $\omega_2$ , and  $\omega_3$  are assigned as 0.2, 0.4, and 0.4, respectively, considering equal importance of Truthfulness and Consistency in controlling hallucination while maintaining prediction accuracy. User alignment received a lower weighting, as it represents less of a limiting factor for advanced models when implemented with optimized prompt engineering. The integrative score was scaled by a factor of 100 to produce normalized values ranging from 0 to 100, enabling straightforward comparison of strategy performance.

## RESULTS AND DISCUSSION

### Requirement for LLM-Based Categorization

Recent advances in HRMS-based nontarget analysis have greatly empowered the discovery of CECs.<sup>2</sup> Nevertheless, the detection of thousands of substances in individual samples (e.g., water, sediment, soil, air) presents growing challenges for robust categorization approaches to support subsequent source apportionment and risk assessment. While structure-based

taxonomic tools, e.g., functional categorization of chemicals through quantitative structure–use relationship (QSUR) models, which infer usage patterns from structural descriptors, are effective for structure-driven predictions, they may not adequately address postidentification decision-support needs that require a better alignment with environmental contextualization. In contrast, our study explored an alternative approach by utilizing an agent-based LLM-assisted framework to perform reasoning-based classification, thereby organizing and contextualizing chemical information derived from textual representations. This framework is particularly useful in postidentification scenarios where chemical identifiers (e.g., IUPAC names or SMILES) have already been available, but the compounds lack contextual annotations such as industrial functions, environmental sources, or risk-relevant classifications and prioritization. As a complementary approach, this LLM pipeline was designed to enrich categorization outputs with information directly relevant to subsequent screening and prioritization works rather than to replace existing structure-based tools.

While databases such as PubChemLite offer broad chemical coverage,<sup>38</sup> NN represents the most comprehensive environmental classification database for regulatory prioritization and monitoring programs, containing over 83,000 substances. Nevertheless, this inventory remains substantially smaller than the estimated chemical exposome of potential concern.<sup>5</sup> For example, more than 350,000 chemicals have been registered for commercial use,<sup>39</sup> and resources such as CCD include over 1.2 million entries. This gap suggests an urgent need to develop an LLM-assisted categorization method to complement NN-based grouping, particularly for CECs not yet listed in NN but likely present in the environment. By applying grouping criteria analogous to those used in NN, we can enhance comparability and support downstream prioritization.

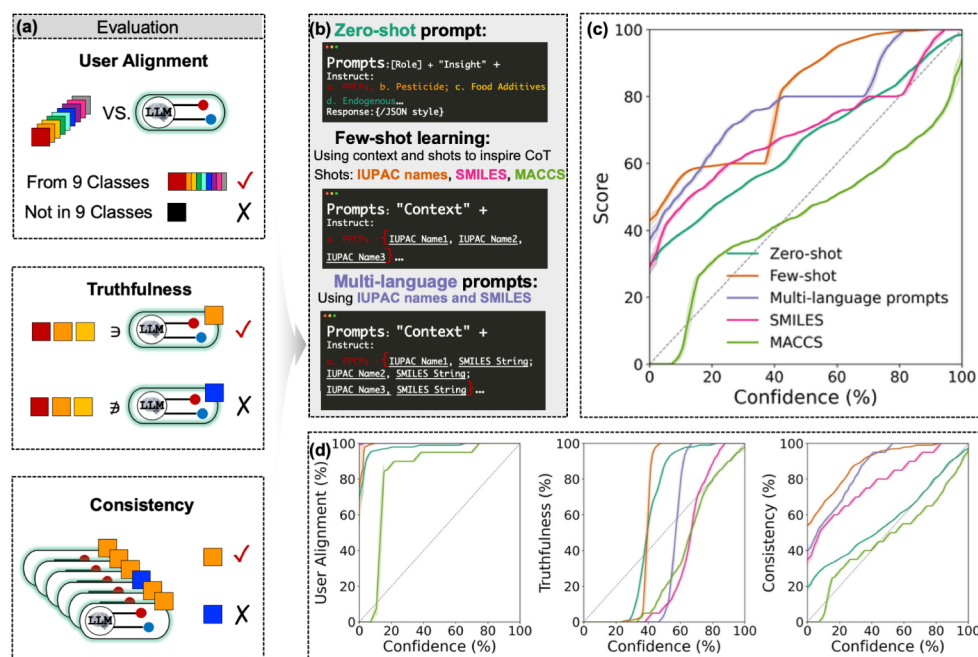
To enable LLMs to accomplish the classification task in a straightforward Q&A manner, the original NN classification system was simplified from 21 to nine broader categories. Five of the 21 categories were removed to reduce noise and improve stability during multiclass reasoning. The remaining categories were reorganized into nine categories based on expert knowledge for the following reasons. First, given that LLMs are built on transformer-based architecture, the attention layer has inherent limitations on input or output tokens.<sup>40</sup> Simplifying the classification scheme ensures that the LLM can retain all task-relevant details without exceeding token thresholds, thus minimizing the risk of losing category definition during subsequent contextual processing.<sup>23</sup> Second, merging semantically analogous categories reduces the likelihood of hallucinations.<sup>41</sup> For instance, the original NN taxonomy distinguishes food additives from food contact chemicals (Figure S1), but such fine distinctions can amplify hallucination risks in semantic reasoning. Using the modified grouping rules, 83,152 substances retrieved from the NN database were assigned to nine predefined categories, achieving 100% coverage of all substances. As shown in Figure S3, the distributions of these categories vary considerably. Notably, endogenous substances account for over 77% of the cataloged entries, despite their limited significance as environmental contaminants. While the NN database is one of the most pertinent resources for environmental studies, this disproportionately high proportion of endogenous substances likely reflects their origins in metabolomics studies. Among typical CECs, PPCPs represent the largest category. Conversely,

metals and industrial intermediates and transformation products are markedly underrepresented, collectively comprising less than 0.5% of the total entries. Although metallic substances are generally easier to identify due to their relatively simple nomenclature, the inadequate coverage of industrial intermediates and transformation products that are among the most common types of CECs significantly reduces the completeness of query results. Furthermore, approximately 20% (16,404 chemicals) of the 83,152 substances had multiple classifications, while the remaining 80% (66,748) were assigned to a single category (Supplementary Sheet 01). Such a high proportion of substances with multicategory assignments may introduce ambiguity in contaminant classification and impede subsequent data interpretation efforts.

Using this predefined categorization framework, we examined an expanded set of 1104 compounds identified by HRMS-based nontarget analysis, including 623 compounds from shale gas FPW and 481 compounds from urban sediment samples (Supplementary Sheet 02). Nowadays, analytical challenges are no longer confined to the detection of contaminants, given recent advances in NTA enabled the effective generation of large chemical inventories from complex environmental matrices. Instead, the bottleneck lies in postidentification interpretation: understanding the environmental significance of these identifiable CECs in terms of functional categories, potential sources, and relevance to risk-oriented prioritization. For mixture risk assessment and management, particularly when the research goal is to acquire a regulatorily practicable list of high-priority substances, a complete chemical list alone is insufficient. Effective prioritization requires contextual information that connects individual compounds to their broader usage patterns, industrial processes, and exposure scenarios across the entire mixture. However, such information is hardly available for a great proportion of NTA-identified contaminants, many of which either are absent from existing environmental classification databases or exhibit ambiguous, multicategory assignments when assessed individually. These limitations revealed a key constraint of conventional categorization strategies that rely on compound-by-compound categorization without contextual awareness. In complex environmental mixtures, especially those dominated by transformation products and industrial intermediates, isolated classification results often fail to synthesize coherent insights regarding the main pollution sources or risk-relevant chemical profiles. This gap highlighted the need for methods capable of integrating classification results across an entire high-risk chemical inventory, providing a structured basis for interpretation and prioritization in subsequent environmental decision-making. AI agent-based frameworks show considerable promise in addressing this integration challenge by enabling mixture-level contextualization, and to ensure that LLMs can reliably support such tasks, it is necessary to systematically validate the performance optimizations achievable within current technical constraints (Supplementary Sheets 05–07).

### Few-Shot Learning Improved Prompts

Although zero-shot prompt (asking straightforward questions without providing the LLM with any examples or response patterns) shows adequate results for general-purpose Q&A scenarios,<sup>42</sup> its limitations become evident when applied to highly professional fields like environmental management, particularly for specialized tasks requiring strict adherence to



**Figure 2.** Comparison of different prompt engineering strategies used in the large language model (LLM)-based classification of contaminants of emerging concern (CECs). (a) The performance of the strategy was evaluated using three metrics: User alignment (UA), Truthfulness (Tr), and Consistency (Con) and summarized using a weighted arithmetic mean to derive an integrative score. (b) Schematic of three prompt types applied for model inference: zero-shot prompt using direct questions, few-shot (individual IUPAC names, SMILES strings, and MACCS fingerprints) prompt including examples and chain-of-thought (CoT) reasoning, and multilanguage prompts combining IUPAC name and SMILES string input. (c) Overall performance comparison of the three prompt strategies, with the integrative score plotted against model confidence. (d) Evaluation of individual performance metrics, indicating improvements by few-shot learning and multilanguage inputs.

classification standards. To address this critical need for accurate categorization of CECs, we adopted a few-shot learning method. To provide examples to help the LLM understand the intent of our task, we followed RLHF principles<sup>26</sup> and compiled nine categories of substances for which the LLM scored full marks (100% correct across 100 trials) during zero-shot evaluation. Three examples (called “shots” in LLM) per category were selected and added to the prompts, providing contextual cues and representative inputs using chemical names or encodings, specifically natural language, chemical language, and computer language names (Figure 2a). The IUPAC names are human-readable systematic nomenclature; SMILES strings encode atom connectivity in linear strings optimized for machine parsing, and MACCS provides binary structural fingerprints indicating substructures. This multiformat prompting aims to minimize misinterpreting chemical inputs. While IUPAC names facilitate symbolic understanding, SMILES strings and MACCS fingerprints offer structural descriptors, enabling LLMs to discern subtle distinctions between functionally or structurally analogous compounds that might lead to classification ambiguity.<sup>43</sup>

Classification performance is shown using cumulative accuracy curves in Figure 2b–c, where each data point corresponds to the percentage of correctly classified substances among the 313 test substances as confidence increases. Confidence here refers to the LLM’s self-assessed probability estimate for correct category assignment. A higher curve indicates that a larger proportion of substances was correctly classified at higher confidence levels. The diagonal dashed line denotes ideal calibration, with curves above this line indicating overperformance and those below implying poor calibration. Comparative analysis of curve characteristics, including vertical

distance from the diagonal line (performance) and horizontal position (confidence) and trajectory steepness, revealed that models exhibiting earlier, steeper ascents while maintaining greater vertical separation demonstrated superior reliability. The results indicate that few-shot prompts significantly outperformed zero-shot across all evaluation dimensions except for MACCS fingerprint inputs, where the improvement was only marginal.

Among all prompt formats, IUPAC names and multilanguage prompts showed the best overall performance (Figure 2b and Supplementary Sheet 04). Multilanguage prompts showed slight superiority at confidence thresholds below 40%, while both formats achieved a Score of 80 at 40% confidence and a full Score at 80% confidence. Notably, both curves displayed pronounced plateau regions, where over 20% of substances persistently scored at intermediate levels (60 or 80), implying challenges in accurately classifying substances with structural ambiguities or those that occupy borderline positions within the classification framework. In contrast, few-shot prompts using SMILES strings showed reduced performance, although they slightly outperformed the zero-shot prompting. MACCS fingerprints with few-shot consistently underperformed compared to other formats, including zero-shot prompting, across the entire confidence range, indicating poor compatibility between MACCS fingerprints and LLM interpretation, suggesting that they are unsuitable as standalone input modalities for this contextual categorization task.

To further determine the optimal use cases for IUPAC-name few-shot versus multilanguage prompts, we analyzed component metrics of the score (Figure 2c and Supplementary Sheets 05–07). Both formats achieved near-perfect User alignment, indicating that the LLM correctly followed instructions and





may produce “Information Cocoons” that compromise performance on niche- or expert-level queries. In additional tests using the latest GPT-5 model, we observed notable improvements in Truthfulness and User alignment, with more than 60% of tested compounds receiving fully correct classifications and 100% accuracy in understanding the context. However, GPT-5 exhibited lower response consistency across repeated runs compared with GPT-4o, indicating that successive model iterations may enhance factual capability while simultaneously increasing variability in responses for specialized domain queries. Among alternative models, Claude-3.5-Sonnet showed the most robust secondary performance, consistent with its documented proficiencies in chemistry-related tasks. GPT-4o and DeepSeek-v3 (671B) exhibited comparable aggregate performances with subtle trade-offs. While DeepSeek-v3 achieved slightly better Truthfulness, it showed reduced Consistency, aligning with recent independent evaluations reporting improved baseline accuracy alongside elevated hallucination frequencies relative to those of GPT-4o. In contrast, Llama-3.1 (405B) showed the weakest classification performance despite its advantages in a lightweight deployment and open-access architecture. It exhibited substantial deficiencies, particularly in User alignment, followed by Truthfulness and Consistency. These limitations illustrated insufficient environmental chemical domain representation during pretraining and suggest that smaller-scale LLMs (including 8B and 70B variants) may require extensive domain adaptation before attaining sufficient reliability for environmental classification tasks.

Although reasoning-optimized models have recently gained attention for outperforming general-purpose LLMs in some Q&A domains via built-in CoT reasoning, their efficacy in domain-specific environmental classification tasks remains uncertain. Two reasoning models, GPT-o1 and DeepSeek-r1 (671B), were compared with their respective general counterparts. As shown in Figures 3c and S10–S15, GPT-4o with a carefully engineered prompt achieved classification scores significantly higher than those of its reasoning-enhanced counterpart GPT-o1. Similarly, DeepSeek-v3 showed comparable superiority over DeepSeek-r1, albeit with a narrower performance gap. Among reasoning models, DeepSeek-r1 outperformed GPT-o1, having higher scores in Truthfulness and Consistency. This suggests that DeepSeek-r1’s reasoning mechanism is more compatible with classification tasks, likely attributable to its structured, multistep semantic decomposition strategy during internal reasoning, which partially emulates prompt engineering logic. These results indicate that current reasoning LLMs remain insufficient for replacing general LLMs augmented with explicit prompt engineering in environmental classification works. This limitation stems largely from their increased susceptibility to hallucination, which compromises reliability despite advanced internal reasoning capabilities.

In summary, model selection for domain-specific tasks should not rely solely on the latest commercial releases (as evidenced by GPT-4o’s underperformance compared to GPT-4-Turbo), nor widely recommended open-source alternatives (e.g., Llama). The present study demonstrated that the effective deployment of generative AI in the environmental management of CECs still requires the integration of human expertise, particularly through sophisticated prompt engineering, to achieve reliable reasoning performance in professional contexts.

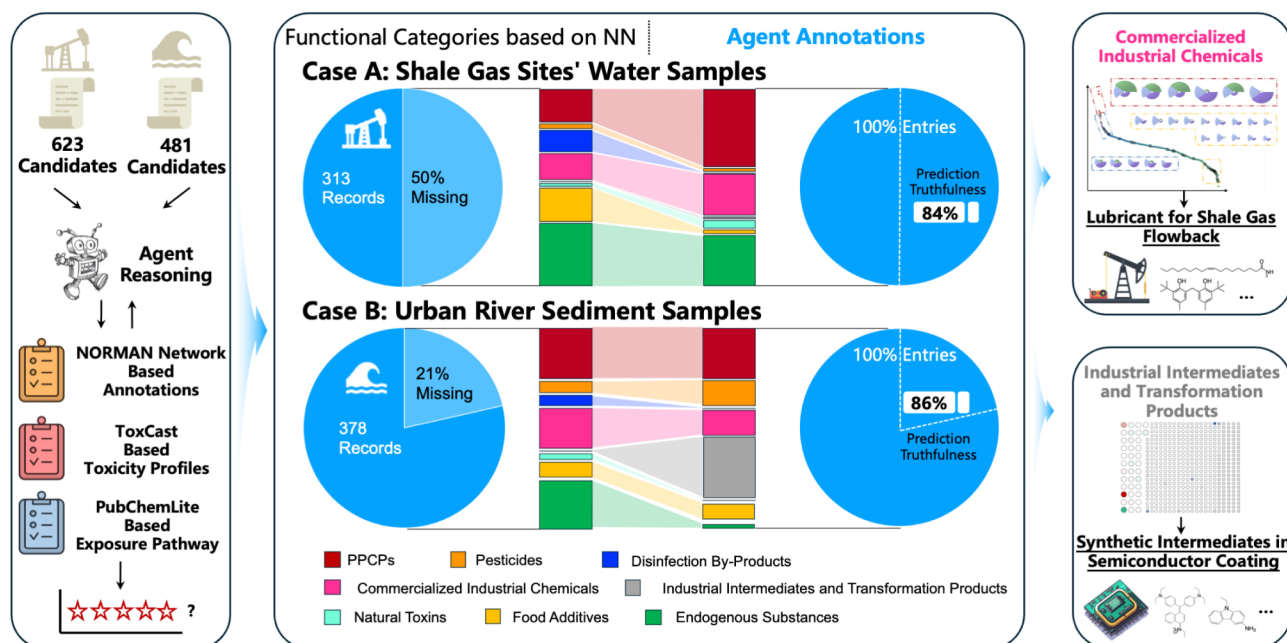
## Knowledge-Augmented and Domain-Tuned LLMs for Improved Classification

When the baseline model performance is inadequate, classification accuracy can be improved by integrating RAG and domain-specific F-T strategies. Our analysis revealed that implementing RAG optimization remarkably enhanced LLMs’ data retrieval capabilities, with noticeable gains in Truthfulness (Figures 3d and S16–S22). The number of substances achieving perfect classification (100% Truthfulness) increased markedly from 170 (54.3%) to 288 (92.0%), while those reaching at least 80% Truthfulness increased from 186 (59.4%) to 299 (95.5%). These improvements indicated that RAG optimization effectively enhanced the model’s reasoning in chemical classification by incorporating a domain-specific knowledge base. Additionally, RAG significantly reduced the odds of misclassification, decreasing the number of substances with 0% Truthfulness from 78 (24.9%) to 7 (2.24%). This enhancement suggests that RAG not only raised classification accuracy but also boosted reliability compared with the original few-shot learning implementation. By enabling a more precise understanding of user intent, accelerating accurate retrieval, and minimizing classification errors, RAG is especially valuable when leveraging publicly accessible databases, allowing more nuanced handling of complex environmental classification challenges.

After RAG optimization, F-T was further applied to enhance the LLM’s contextual understanding of environmental chemicals. The LLM achieved a relatively high level of Truthfulness following RAG optimization, with F-T providing only marginal gains in classification accuracy (Figures 3d and S16–S22). This implied that RAG’s integration of domain-specific knowledge is almost sufficient for achieving high-precision classification, reducing the relative value of computationally intensive F-T for accuracy enhancement. In contrast, F-T contributed significantly to output consistency, increasing the number of substances with 100% Consistency from 98 (31.3%) using RAG alone to 190 (60.7%) with F-T being added, and those with 90% Consistency from 150 (47.9%) to 276 (88.2%). This marked improvement in reproducibility highlights F-T’s role in refining the model’s professional contextualization, effectively reducing response variability, and bolstering operational stability.

Beyond RAG and F-T, agent-based implementation further improved the overall classification performance by enforcing workflow-level constraints and coordinating evidence retrieval. As shown in Figures 3d and S16–S22, all agent-based classification results achieved integrative Scores above 80, with Truthfulness exceeding 97% across the test substances and User alignment reaching 100%, indicating strict adherence to task instructions and output formats. Compared with the baseline GPT-4-Turbo and RAG-enhanced configurations, the agent implementation significantly increased the averaged integrative Score ( $p < 0.01$ ), reflecting a systematic rather than isolated performance gain. However, improvements in Consistency were limited. Although the agent slightly outperformed GPT-4-Turbo in repeated query stability, the difference was not statistically significant. These results indicate that while agent-based coordination can enhance correctness and robustness at the workflow level, hallucination-related variability remains an inherent limitation of current LLMs that cannot be fully resolved through agent design alone. A similar pattern was observed when the workflow was evaluated using the GPT-5 model series. Agent configurations





**Figure 4.** Agent-based decision support for post-prioritization of contaminants in environmental samples. The agent integrates NORMAN Network (NN)-based functional categorization with structured toxicological (ToxCast) and exposure-related (PubChemLite) information to support decision-making after screening. Two application scenarios are shown: shale gas site water samples and urban river sediment samples. By embedding LLMs, information gaps affecting approximately 50% and 21% of detected chemicals under unclear annotation were supplemented through multisource evidence integration. Manual checks confirmed 84% and 86% prediction truthfulness, respectively, in the context of the NN database. Based on scenario-specific categorization, lubricants used in flowback were identified as emerging high-risk contributors at shale gas extraction sites, while urban river sediments from the Pearl River Delta were dominated by industrial intermediate compounds associated with semiconductor dye synthesis that exhibit elevated aquatic ecological risks. The resulting risk ranking patterns are reprinted in refs. 35 and 36.

again achieved the highest integrative Score, while the fine-tuned models showed performance comparable to the RAG-enhanced setup, with both approaches substantially outperforming the corresponding baseline model. Nevertheless, the overall performance yielded with GPT-5 remained slightly lower than that achieved with GPT-4-Turbo in this work. The result suggested that although agent orchestration, RAG, and fine-tuning can systematically improve task-specific performance, the ultimate effectiveness of these strategies still depends on the capabilities and optimization characteristics of the underlying baseline model.

#### Agent-Guided Interpretation of Chemical Mixtures in Environmental Samples

Within the agent-based framework, categorization outputs serve as an organizing layer to support downstream interpretation of complete chemical inventories rather than as standalone analytical results. Accordingly, our analysis showed aggregated patterns across high-priority substance lists, focusing on how coordinated classification and contextual interpretation can recognize the main functional groups and scenario-specific risk contributors. This approach allows the extraction of mixture-level signals from large datasets of identified CECs, which cannot be discerned by isolated, compound-by-compound inspection, particularly for complex mixtures dominated by commercialized industrial chemicals and their intermediates and transformation products. Moreover, automated methods that integrate appropriately designed AI agents offer a promising solution toward efficient, rapid, and more accurate interpretation. Our study presents the first attempt to develop such an agent-based strategy to address environmental challenges associated with chemical mixtures.

To assess whether LLMs can compensate for dataset bias through reasoning and contextual understanding, we selected shale gas FPW and urban sediments as two real-world scenarios that represent distinct environmental media and chemical profiles. As shown in Figure 4, the classification patterns in the two scenarios differed markedly, reflecting distinct primary chemical sources and usage contexts.

At shale gas extraction sites, this mixture-level interpretation revealed a heterogeneous classification profile dominated by PPCPs, endogenous substances, and commercialized industrial chemicals. The frequent occurrence of PPCPs and endogenous compounds, though initially unexpected, can be reasonably explained by their extensive use and widespread environmental distribution in surrounding environments, which may enter extraction sites, and in situ biogeochemical processes may also contribute endogenous metabolites. Alternatively, commercialized industrial chemicals emerged as the most informative category for risk interpretation in these FPW samples, as their occurrence is more directly tied to site-specific activities rather than background contamination. Notably, within this category, our previous NTA work<sup>35</sup> identified six high-risk compounds that had never been reported in shale gas extraction sites and are not included in the U.S. Environmental Protection Agency's list of 1173 chemicals (<https://www.epa.gov/hfstudy>). Four of these substances appear unique to Chinese sites. Through agent-based contextual annotation, these compounds were linked to lubricants often used in shale gas flowback operations, offering a plausible explanation for their source. Subsequent review and comparison with on-site material inventories supported this inference, confirming lubricants as a previously underappreciated source of high-risk contaminants in shale gas FPW.

Interpreting NTA data of urban sediment samples from the Pearl River Delta revealed a distinctly different mixture structure, characterized by pronounced shifts in category composition following knowledge augmentation. These shifts indicated that the changes arise from the incorporation of newly inferred information rather than from modifications to the underlying chemical inventory. For PPCPs, category proportions remained largely unchanged before and after agent-based annotation, consistent with the fact that PPCPs in urban aquatic environments have already been well documented in existing databases, and thus, annotation provided limited added value. In contrast, transformation products and industrial intermediates showed significantly improved representativeness following an agent-based interpretation. This increase demonstrated that such substances have long been systematically underrepresented in conventional environmental databases, with a large number of newly annotated compounds emerging from LLM-assisted inference. Manual verification confirmed that over 86% of these inferred classifications were correct, indicating that despite the limited coverage of transformation products in existing inventories, they can be meaningfully contextualized through our developed LLM-based reasoning framework, thereby complementing current databases. Building on our previous work that identified 390 compounds exceeding risk thresholds through combined quantitative and semiquantitative assessment,<sup>36</sup> a key unresolved question persisted. Do these high-risk substances share identifiable commonalities that could inform upstream chemical management or downstream control strategies? Through mixture-level interpretation of this prioritized chemical dataset, the present study revealed a pronounced pattern. The vast majority of industry-sourced compounds with risk quotients exceeding 0.1 were linked to intermediates used in the synthesis of semiconductor spray-coating dyes. This convergence remained invisible when compounds were examined individually but emerged clearly when classification results were aggregated and interpreted at the mixture level using agent-based interpretation. These findings highlighted how agent-based interpretation can transform extensive CECs outputs into compressed, source-oriented risk narratives, thereby delivering actionable information for environmental chemical management.

### Current Limitations and Future Perspectives

While promising, LLM-based contextual interpretation still faces challenges. Notably, while LLMs show great potential for inferring the functional context of transformation products and industrial intermediates, their performance is constrained by the limited coverage of such compounds in available environmental knowledge bases. When transformation pathways, precursor-product relationships, or usage contexts are poorly documented, LLMs may struggle to generate fully informed interpretations, even with optimized prompting or retrieval settings. This limitation highlights the importance of continuously expanding and curating domain-specific knowledge resources. The agent-based framework implemented in this study explicitly provides APIs for knowledge supplementation of RAG, allowing dynamic integration of external databases, expert-curated rules, and newly generated annotations to support iterative refinement of classification and interpretation results.

Although F-T helps alleviate data imbalance to some extent, model performance is strongly dependent on the comprehen-

siveness and quality of available datasets. Incomplete or biased F-T data would limit the classification accuracy. Recent advances, such as data-distillation techniques (e.g., DeepSeek), offer cost-efficient alternatives for transferring task-specific knowledge from large LLMs to smaller models, bypassing the need for full retraining. Moreover, hallucinations, where the model generates incorrect or unsupported information, remain a challenge, even after optimization. Future improvements may include deeper domain-specific F-T and possibly tuning an LLM from scratch with a focus on environmental data, which may reduce hallucinations and improve reliability. It should also be noted that both the reasoning and predictive datasets used in the current study originated from the same environmental source, the shale gas FPW. While this design allowed us to simulate a realistic and chemically complex postidentification scenario, it may introduce potential information leakage and limit the generalizability of the results. As such, the applicability of the proposed LLM-based classification framework to samples from other environmental contexts (e.g., municipal wastewater, industrial discharges, and surface waters) remains to be validated. Additional dataset related to urban sediments confirmed the applicability of the developed framework. Future work should extend this pipeline to chemically and geographically diverse datasets to further evaluate its robustness, domain transferability, and real-world utility in broader environmental monitoring applications.

Future applications such as predicting chemical transformations and environmental fate may require multimodal LLM architectures or agent-based systems that integrate with structure-driven models to support simulation-based reasoning. Such advances may pave the way for a new generation of LLMs optimized for environmental studies, such as pretraining foundation models with the literature and database, allowing for a more effective, accurate, and comprehensive understanding of chemical mixtures in the environment.

## ■ ASSOCIATED CONTENT

### Data Availability Statement

The optimized agent system has been made publicly available to support local deployment. The files can be accessed via the Hugging Face repository at [https://huggingface.co/Qianhui19/CECs\\_annotating\\_agent](https://huggingface.co/Qianhui19/CECs_annotating_agent). In the meantime, an online demo version is also accessible as a published ChatGPT application at <https://chatgpt.com/g/g-6762794190648191b24f54d1140bfc8e-cecmix-agent>.

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.6c01342>.

Text describing detailed RAG and F-T procedures and prompts; figures showing predefined classification rules used for the GPT and benchmarking performance (PDF)

Supplementary sheets detailing chemical lists and evaluation results (XLSX)

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## Notes

The authors declare no competing financial interest.

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